

MEXAN TABLET 80 MG

Composition

A white round biplane tablet with the cross-mark on one side and letter M on the other side. Each tablet contains gliclazide 80 mg.

Properties

Gliclazide is a second generation sulphonylurea oral hypoglycaemic agent used in the treatment of non-insulin-dependent diabetes mellitus (NIDDM). Gliclazide reduces blood glucose levels in healthy volunteers and patients with NIDDM by correcting both defective insulin secretion and peripheral insulin resistance. Unstimulated and stimulated insulin secretion from pancreatic β -cells is increased following administration of gliclazide, with both the first and second phases of secretion being affected. This occurs via the binding of gliclazide to specific receptors on pancreatic β -cells, which results in a decrease in potassium efflux and causes depolarisation of the cell. Subsequently, calcium channels open, leading to an increase in intracellular calcium and induction of insulin release. In addition, gliclazide increases the sensitivity of β -cells to glucose.

Gliclazide may have extrapancreatic effects which restore peripheral insulin sensitivity, such as decreasing hepatic glucose production and increasing glucose clearance and skeletal muscle glycogen synthase activity. These effects do not appear to be mediated by an effect on insulin receptor number, affinity or function.

Administration of a single oral dose of gliclazide 40 to 120 mg in healthy volunteers results in a maximum serum concentration C_{max} of 2.2 to 8 mg/L within 2 to 8 hours. Both the C_{max} and t_{max} are increased after repeated gliclazide administration, although drug accumulation has not been observed. Steady-state concentrations are achieved after 2 days' administration of gliclazide 40 to 120mg. Administration of gliclazide with food reduces C_{max} and delay t_{max} .

The volume of distribution of gliclazide is low (13 – 24L) and the drug is extensively protein-bound (85 – 97%). The elimination half-life is 8.1 to 20.5 hours. Gliclazide is extensively metabolised to 7 metabolites which are predominantly excreted in the urine. 60 – 70% of the dose is excreted in the urine and 10 – 20% in the faeces.

Indications

Treatment of maturity onset diabetes mellitus.

Contraindications

Mexan should not be used in:

- Juvenile onset diabetes
- Diabetes complicated by ketosis and acidosis.
- Pregnancy
- Diabetes undergoing surgery, after severe trauma or during infections.
- Diabetic pre-coma and coma
- Severe renal or hepatic insufficiency
- Patients hypersensitive to sulphonylureas and related drugs.

Side effects

Hypoglycaemia - symptoms include sweating, dizziness, tremor, visual disturbances, intense hunger, slurred speech, weakness, poor concentration and confusion.

Gastrointestinal disturbance include nausea, dyspepsia, diarrhoea, constipation, vomiting have been reported. The GI disturbances can be minimised if gliclazide is taken during a meal.

Skin reactions such as rash, pruritus, erythema, flushing and bullous eruption have also been reported.

Abnormality of hepatic function with raised ALT and AST levels may occur during gliclazide therapy. There have also been rare reports of hepatic failure, hepatitis and jaundice following treatment with gliclazide. Rare reports of blood dyscrasia including anaemia, leucopenia, thrombocytopenia and granulocytopenia have also been associated with use of gliclazide.

Precautions / Warning

Hypoglycaemia: All sulphonylurea drugs are capable of producing moderate to severe hypoglycaemia, particularly in patients controlled by diet alone; in cases of accidental overdose; in cases where calories or glucose intake is deficient; in patients with hepatic and/or renal impairment. However, in long-term clinical trials, patients with renal insufficiency have been treated satisfactorily with gliclazide at reduced doses. In order to reduce risk of hypoglycaemia, it is recommended:

- to initiate treatment for non-insulin dependent diabetes by diet alone if possible;
- to take into account the age of the patient: blood sugar levels not strictly controlled by diet alone might be acceptable in the elderly;
- to adjust the dose of Mexan according to the blood glucose response and to the 24 hours urinary glucose during the first days of treatment.

Dosage adjustment may be necessary:

- on the occurrence of mild symptoms of hypoglycaemia (sweating, pallor, hunger pangs, tachycardia, sensation of malaise). Such findings should be treated with oral glucose and adjustment made in drug dosage and/or meal patterns.
- On the occurrence of severe hypoglycaemia reactions (coma or neurological impairment);
- Loss of control of blood glucose (hyperglycaemia).

When a patient stabilised on any diabetic regimen is exposed to stress such as fever, trauma, infection or surgery, a loss of control may occur. At such times, it may be necessary to progressively increase the dosage of Mexan and if this is sufficient, to discontinue the treatment with the oral sulphonylurea and to administer insulin.

Care should be exercised in patients with hepatic and/or renal impairment. A small starting dose should be used with careful monitoring. As with other sulphonylureas, hypoglycaemia will occur if the patients' dietary intake is reduced or if they are receiving a larger dose of Mexan than required.

Effects on ability to drive and use machines: Patients should be informed that their concentration might be affected if their diabetes is not satisfactory controlled, especially at the beginning of treatment.

Use in Pregnancy and Lactation:

See Contraindication. Sulphonylureas have been found to be mildly fetotoxic in rat reproductive studies. There are no adequate and well controlled studies in pregnant women.

Because recent information suggests that abnormal blood glucose levels during pregnancy are associated with a higher incidence of congenital abnormalities, many experts recommend that insulin should be used during pregnancy to maintain blood glucose levels as close to normal as possible.

Nursing mother: It has not yet been established whether gliclazide is excreted into human milk. However, other sulphonylureas have been found in milk and there is no evidence to suggest that gliclazide differs from the group in this respect.

Use in children

Mexan, like other sulphonylureas, is not indicated for the treatment of juvenile onset diabetes mellitus.

Drug interactions

Care should be exercised when giving Mexan concomitantly with drugs known to alter the diabetic state or potentiate the effects of gliclazide.

Drugs which may potentiate hypoglycaemia effect by increase in plasma gliclazide concentrations: sulphonamides, phenylbutazone, salicylates, clofibrate, allopurinol, coumarin derivatives, miconazole, disopyramide, vitamin K antagonists, cimetidine.

Drugs which may potentiate hypoglycaemic effect by stimulating insulin secretion or suppressing manifestations of hypoglycaemia: theophylline, caffeine, MAO inhibitors, beta adrenergic antagonists.

Drugs which may reduce the hypoglycaemic effects of reducing plasma gliclazide concentrations: barbiturates, phenytoin, rifampicin.

Drugs which may increase blood glucose levels resulting in reduction in glycaemic control: corticosteroids, oral contraceptives, oestrogens, progestogens, phenytoin.

Dosage and Administration

Adult: The total daily dose may vary from 40 to 320mg taken orally. Dosage should be adjusted according to individual patients' response, commencing with 40 – 80mg daily, and increasing until adequate control is achieved. A single dose should not exceed 160mg. When higher doses are required, Mexan should be taken twice daily and according to the main meals of the day. In obese patients or those not showing adequate response to Mexan alone, additional therapy may be required.

Elderly: Plasma clearance of gliclazide is not altered in the elderly and steady state plasma levels can therefore be expected to be similar to those in adults under 65 years. Clinical experience in the elderly to date shows that gliclazide is effective and well tolerated. Care should be exercised, however, when prescribing sulphonylureas in the elderly due to a possible age-related risk of hypoglycaemia.

Children: not recommended.

Overdosage

The symptoms to be expected with an overdose would be hypoglycaemia. The treatment is gastric lavage and correction of hypoglycaemia by appropriate means with continued monitoring of the patients' blood sugar until the effect of the drug has ceased.

Packing

Box of 10 X 10's and 50 x 10's tablets.

Storage

Store in a cool dry place, below 30°C. Protect from light.

Shelf-life: Please refer to the outer box label. Keep out of reach of children

Jauhi daripada kanak-kanak.

Manufactured by:

T.O. CHEMICALS (1979) LTD.

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